# Tushar Sharma

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## INTRODUCTION

Passionate and self-motivated Computer Science student with hands-on experience in Data Science and Machine Learning. Skilled in Python, SQL, and deep learning frameworks with practical expertise and end-to-end projects. Actively exploring advanced technologies such as Artificial Intelligence and Quantum Computing. Eager to contribute analytical, problem-solving, and engineering skills to real-world data-driven challenges in dynamic environments.

## EDUCATION

**Adarsh Madhyamik Vidyalaya** **2019-2021**
*Higher Secondary-* ***PERC - 75.6*** *Kolkata,West Bengal*

**Sister Nivedita University** **2022 – 2026**
*B.Tech - Computer Science and Engineering -* ***CGPA - 8.50*** *Kolkata,West Bengal*

## TECHNICAL SKILLS

**Languages:** Python, C/C++, JavaScript, Java, SQL
**Technologies/Frameworks:** TensorFlow, Scikit-learn, LangChain, Pinecone, Streamlit, Cloudinary API,Gmail API Node.js, JWT, Qiskit
**Machine Learning / AI Skills:** RAG Models, Sentence Transformers, Word Embeddings, NLP, Vector Databases, Feature Engineering,Quantum Machine Learning
**Developer Tools:** VSCode, Jupyter, Google Colab, IBM Quantum Platform, Git/GitHub

## RESEARCH PAPER

**A Quantum-Driven Probabilistic Feature Extraction Approach for Tumor Classification**

*Accepted & Presented — Quantum Kol ,Springer, 2025*

- **Developed a Quantum-Inspired Framework** – Proposed a probabilistic feature extraction method for MRI-based tumor classification using quantum principles.
- **Integrated Quantum and Classical Techniques** – Combined quantum image representation with classical **SVM**, achieving **80% accuracy** on Br35H dataset.
- **Enhanced Diagnostic Precision** – Demonstrated the capability of **Quantum Machine Learning (QML)** to improve efficiency and accuracy in medical imaging analysis.

## EXPERIENCE

**Data Science Internship at IDEAS, ISI Kolkata** **January 2025 – April 2025**
*Role : TIH Intern* *Kolkata, West Bengal*

- **Automatic Data Extraction & Training:** Extracted structured and unstructured data from 100+ page PDFs under **Tapas Chakrabarty (Vixplor Analytics)**, enhancing **data accessibility** and minimizing **manual effort**. Completed **32-hour training** on Python for Data Science, Data Visualization, Streamlit, Codeless ML tools, and Deep Learning basics.
- **Skills & Outcomes:** Gained expertise in **Data Preprocessing**, **Feature Engineering**, **Python**, **Visualization**, and **ML Fundamentals**, strengthening **Problem-Solving** and **Analytical Thinking**.

**Machine Learning Internship CodSoft** **September 2025 – October 2025**
**Handwritten Text Synthesis using LSTM** | Deep Learning / Sequence Modeling *Remote*

- **LSTM-Based Handwriting Generation System:** Developed a deep learning model for **handwritten text synthesis** using a **stacked LSTM architecture** with **attention-based windowing**. Modeled handwriting as sequential pen-stroke data ($\Delta x, \Delta y$, pen-up/pen-down) conditioned on input character sequences, enabling realistic and continuous handwriting generation without explicit character segmentation.

- **Skills & Outcomes:** Successfully generated smooth and human-like handwriting with correct character alignment, validated through **attention weight visualizations** showing monotonic text–stroke correspondence. Strengthened expertise in **Recurrent Neural Networks (LSTM)**, **Sequence-to-Sequence Modeling**, **Attention Mechanisms**, **Probabilistic Output Modeling**, and **Deep Learning using Python**.

## PROJECTS

**Medical Assistant Chat System** | AI-Powered RAG Application **Oct 2025**

- **AI-Powered Chat System:** Built a **Retrieval-Augmented Generation (RAG)** model for answering medical queries from PDFs using **LLM-based contextual retrieval**. Implemented **Vector Search** with **all-MiniLM-L6-v2**, **Pinecone**, and **LangChain**, and developed an interactive **Streamlit app** integrated with **ChatGroq** and **LLaMA-3.3-70B-Versatile**.
- **Skills & Outcomes:** Gained expertise in **LLM Integration**, **RAG Pipelines**, **Vector Databases**, and **Prompt Engineering**. Enhanced knowledge in **Streamlit Web Development**, **LangChain Framework**, and deploying **AI-driven conversational systems**.

**Multiclass Hepatic Tumor & Vessel Segmentation** | Deep Learning / Medical Imaging **Dec 2025**

- **Swin-UNet–Based Segmentation System:** Designed and implemented a **Transformer-based Swin-UNet architecture** for automated multiclass segmentation of **hepatic tumors and blood vessels** from CT/MRI images. Leveraged **shifted-window self-attention** to capture long-range spatial dependencies and a **U-Net–style encoder–decoder** with skip connections for precise boundary localization. Trained the model using a **Dice + Cross-Entropy loss** to effectively handle class imbalance.
- **Skills & Outcomes:** Achieved **Tumor Dice score of 89.42%** and **Vessel Dice score of 83.06%**, outperforming baseline U-Net models. Strengthened expertise in **Medical Image Segmentation**, **Vision Transformers**, **PyTorch**, and **Model Evaluation Metrics** (Dice, Precision, Recall, HD95).

## CERTIFICATIONS

- Deep Learning For Developers-Infosys
- Applied Machine Learning in Python-Coursera
- Quantum Computing-CDAC & IIT Roorkee
- Develop Generative AI Applications-IBM skill Network